



Problem Analysis



FOOD FOR THOUGHT
CHALLENGE

➡ Redefined Problem Statement

- College students crave **quick, affordable** and **tasty** late-night food but existing options like Swiggy delivery or hostel canteens **fail to meet their needs** for timing, budget, and convenience.
- The main issue is not hunger. It is the absence of an easily accessible, low-cost, and round-the-clock food option that suits the student lifestyle on campus.

➡ Students Needs

➡ Late-Night Food Access

- Most students study or socialize until 2 or 3 AM. Food outlets and canteens close by 10 PM, leaving no good options on campus.

➡ Affordable Meals

- Students usually spend ₹80 to ₹150 per meal. However, late-night orders from Swiggy or Zomato cost ₹250 to ₹300 or more, which includes delivery and packaging.

➡ Quick Availability

- They want food ready within 8 - 10 minutes, without walking to the gate or waiting for delivery.

➡ Student Challenges

➡ Gate Restrictions

- At IITs and other universities, security gates close by 10 PM to 11 PM. This means riders cannot enter, and students have to walk long distances to get food.

➡ Limited Late-Night Options

- Hostel canteens close early, and only a few nearby restaurants stay open past midnight. This is especially true in Tier 2 and Tier 3 cities where most campuses are located.

➡ High Delivery Fees

- Late-night orders come with extra charges of ₹70 to ₹100. This makes ordering through apps too expensive for students.

➡ Is late-night food demand increasing?

- Swiggy reported a **23% surge** in orders **after 11 pm**, making late-night orders ~8.5% of total orders.

➡ Are there barriers to late-night food access for students?

- At IITs and other universities, security gates **close by 10 PM to 11 PM**. This means riders cannot enter, and students have to walk long distances to get food.

Source

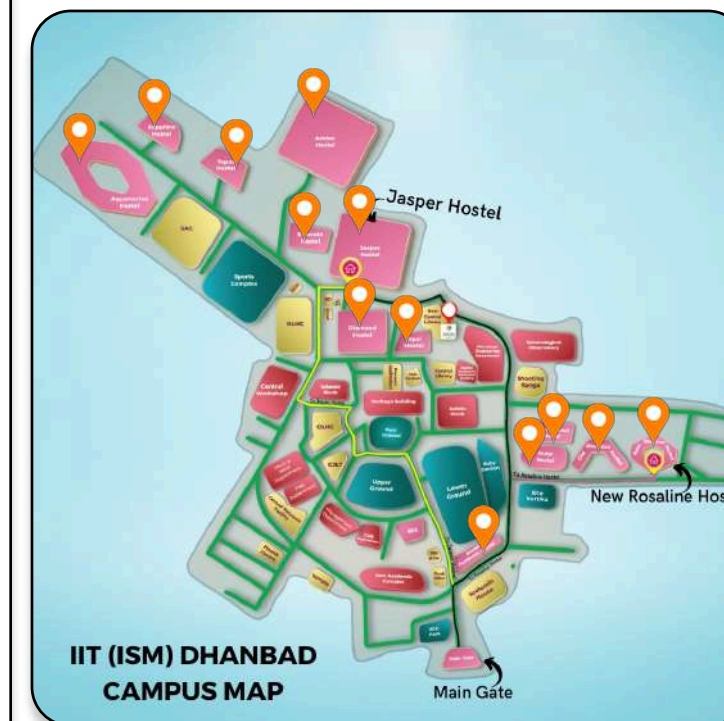
➡ Market Potential (From One Hostel to All of India)

Aquamarine Hostel (Night Canteen Data)

- Hostel capacity: ~2,000 students
- Based on real night canteen data: ~400 daily orders placed after 10 PM
- That means around 20% of students order food nightly
- Average spend per order: ₹70
- Daily revenue: ₹28,000
- Monthly potential: ₹8.4 Lakh
- Annual potential: ~₹1 Crore

Entire IIT (ISM) Dhanbad

- Total students: ~6,500 (99% reside in hostels)
- Following the same trend, ~20% students = 1,300 daily night-food users
- Avg. order value: ₹70
- Daily revenue: ₹91,000
- Monthly: ₹27 Lakh
- Annual potential: ₹3-3.5 Crore

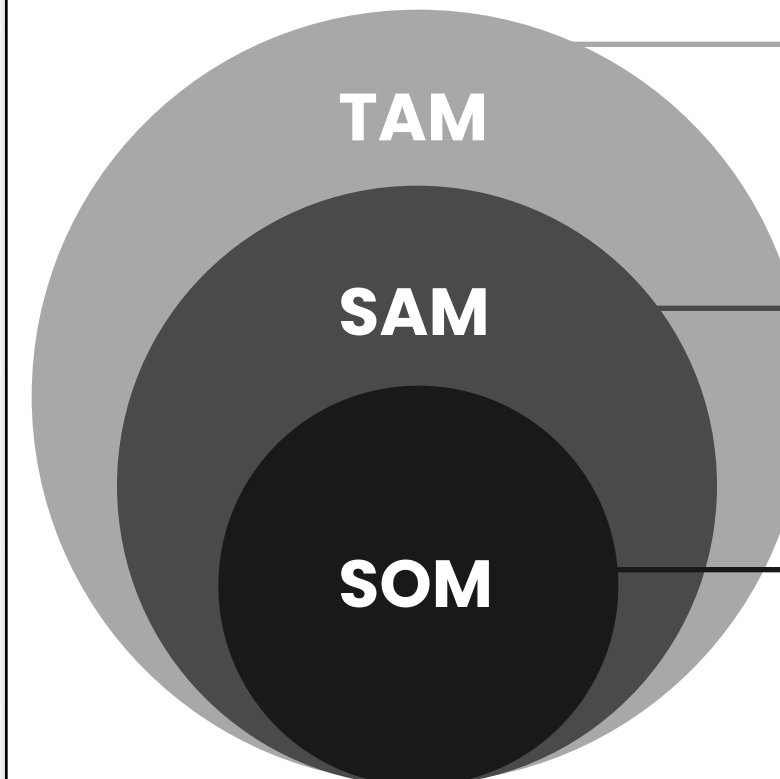


All 23 IITs (Scaled Estimate)

- Avg. 6,000 students × 23 = ~1.38 Lakh total students
- 20% active nightly → 27,600 daily users
- ₹70/order → ₹19 Lakh/day
- Monthly potential: ₹5.8 Crore
- Annual potential: ₹70+ Crore

All Across India (Hostel Student Market)

- India has ~3.7 million hostel students (Ministry of Education, 2023)
- Assuming similar behavior → 20% order nightly = 7.4 lakh daily users
- ₹70/order → ₹5.18 Crore/day
- Monthly: ₹155 Crore
- Annual market potential: **₹1,800-₹2,000 Crore**



TAM: India's late-night hostel food market is expanding rapidly, driven by 3.7 million students living in hostels across the country.

3.7M hostel students × ₹70 avg. order × 20% daily users × 365 days = **~₹2,000 Crore/year**

SAM: Tier-1 & Tier-2 colleges (IITs, NITs, IIITs, IIMs, and top private universities) represent the most accessible and digitally active 25% of this market.

₹2,000 Cr × 25% = **~₹500 Crore/year**

SOM: The immediate focus area-all 23 IIT campuses, where 1.38 lakh students generate consistent night-time food demand.

1.38L students × ₹70/order × 20% daily users × 365 days = **~₹70 Crore/year**



Hydro Vending



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What is HydroVending ?

- **HydroVending** is an automated smart vending machine that serves **fresh, hot meals** to students **anytime** within their campus.
- It uses **vacuum dehydration technology** to rehydrate pre-cooked dry meals with precise hot water and automatic stirring.
- The machine ensures **quick, affordable, and convenient** food access right **inside hostels**, without needing any delivery or gate entry.



How does HydroVending achieve all these values?

→ Late-Night Availability



- Operates **24x7 inside hostels** and study zones, ensuring students always have access to food when needed.
- Provides fresh, hot meals even after canteens close, solving the major problem of late-night hunger on campus.

→ Affordability



- Keeps average meal prices under ₹100, making it pocket-friendly and affordable for every student on campus.
- Uses vacuum-dried meal technology to minimize preparation cost while maintaining great taste and food quality.

→ On-Hostel Access



- Machines are installed directly inside hostels, removing gate restrictions and delivery dependency during late hours.
- Students can enjoy freshly prepared meals within their hostel without worrying about delivery delays or campus security rules.

→ Quick & Convenient Access



- Prepares hot meals in just 2–3 minutes, eliminating long delivery waits or the need to step outside.
- Integrated touchscreen and QR-based payment system make the ordering and collection process smooth and effortless.

Technology behind HydroVending

1 Food Preparation & Standardization

- All meals are prepared in a controlled central kitchen using industrial cookers and **precise recipe ratios**.
- Each dish is batch-tested for flavor, texture, and moisture balance before moving to the dehydration stage.

3 Smart Packaging & Sealing

- Dehydrated food is sealed in laminated pouches using **nitrogen flushing** to maintain quality and avoid oxidation.
- Each pouch includes a fixed **desiccant pocket** and QR code for batch traceability and machine compatibility.

4 Machine Loading & Monitoring

- Pouches are loaded into HydroVending units installed inside hostel buildings for **direct student access**.
- Every machine uses **IoT sensors** and **cloud dashboards** to track temperature, stock levels, and cleaning cycles.

2 Vacuum Dehydration (Core Technology)

- In this step, food is placed in a **vacuum chamber**, where moisture is removed at low pressure and temperature.
- Unlike normal hot-air drying, this process **prevents nutrient loss** and retains original flavor and structure of food.
- The final product becomes **light, crisp, and shelf-stable**, ready to regain freshness when hot water is added.



5 Hot Meal Ready in Minutes

- When an order is placed, the system dispenses the dry mix and injects precise **95°C hot water** automatically.
- A microcontroller regulates **heating, timing, and stirring**, ensuring uniform rehydration and perfect texture every time.
- In just **2–3 minutes**, the machine delivers a fresh, steaming bowl of food identical to the original cooked meal.





Dining Out



FOOD FOR THOUGHT CHALLENGE

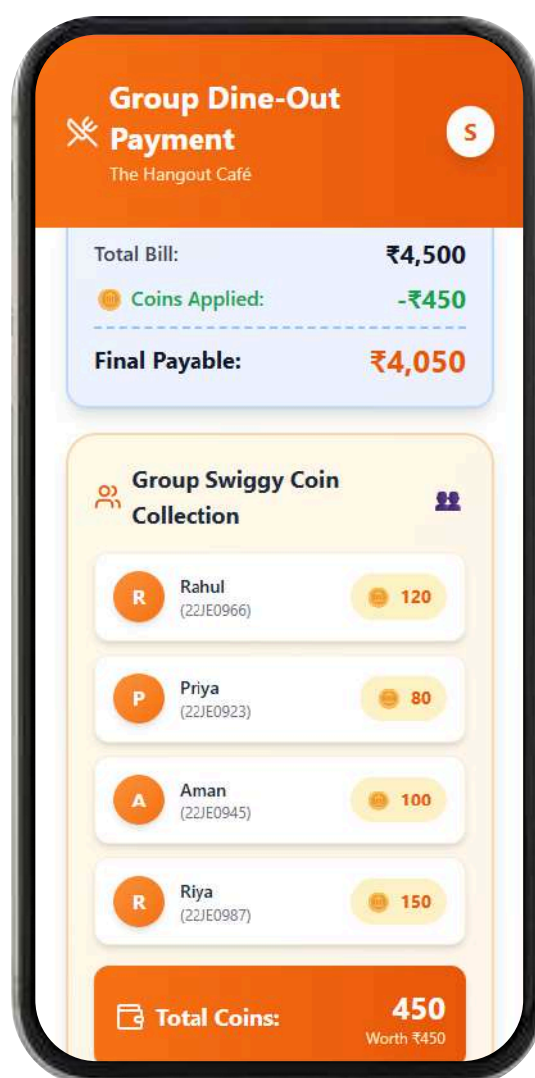
Why Students Think Twice Before Dining Out ?

- Students usually plan dine-outs for **Birthday Party's, Celebrations, or Group Treats** with close friends.
- But a single dine-out for 4–6 friends often costs **around ₹4,000–₹5,000**.
- A small **₹200 voucher** barely reduces the expense, so it feels unworthy of effort.

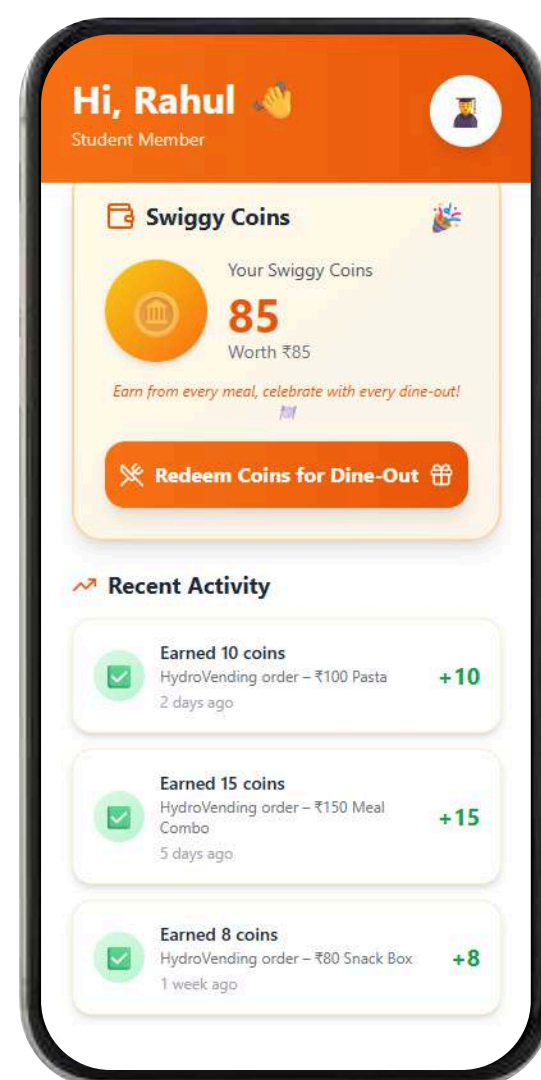


How Swiggy Coins Make Dining Out More Affordable

- Students earn Swiggy **Coins worth 10%** every time they order meals from HydroVending machines.
- All earned coins are automatically added to each **student's Swiggy account** inside the app wallet.



- During a group dine-out, friends can **combine their collected Swiggy Coins** directly through the app.
- The total **collected coins** are applied instantly during payment to reduce the overall restaurant bill.
- By pooling coins, a group can easily save **₹300–₹500** on a **₹4,000–₹5,000 dine-out**.

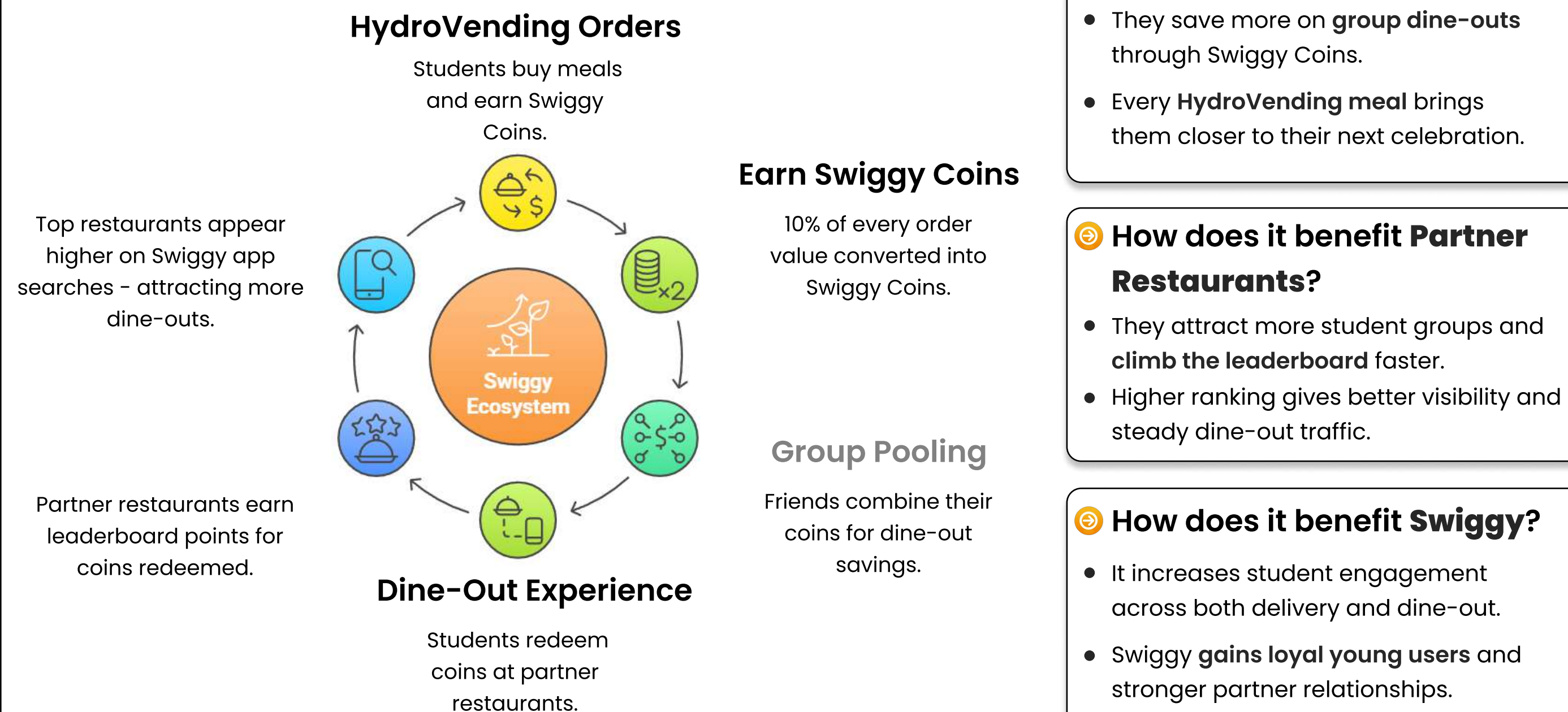


What is the Swiggy Coins Leaderboard?

- Ranks partner restaurants **region-wise** every month based on **total Swiggy Coins collected** from student dine-outs.
- Encourages healthy competition - **motivating restaurants** to offer better food, service, and student experiences to climb the ranks.
- Top 3** restaurants get featured on the Swiggy app, **appearing first in local searches**.

Swiggy Swiggy Coins Leaderboard — Dhanbad Region		
Monthly ranking of partner restaurants by total Swiggy Coins collected from student dine-outs.		
Top 3 restaurants get priority visibility on Swiggy app search results		
October 2025 Rankings		
1	The Hangout Café North Campus	₹12,400 coins
2	Pizza Junction City Centre	₹9,800 coins
3	Campus Cantina South Campus	₹8,900 coins
4	The Late Bite Tech Park	₹7,200 coins
5	Coffee House 24x7 Main Market	₹6,800 coins
Leaderboard Info		Updated: Monthly Current Period: October 2025 Region: Dhanbad Total Partners: 124 Restaurants
Join the Leaderboard Become a Swiggy Partner		

The Swiggy Campus Ecosystem Loop



How does it benefit Students?

- They save more on **group dine-outs** through Swiggy Coins.
- Every **HydroVending meal** brings them closer to their next celebration.

How does it benefit Partner Restaurants?

- They attract more student groups and **climb the leaderboard** faster.
- Higher ranking gives better visibility and steady dine-out traffic.

How does it benefit Swiggy?

- It increases student engagement across both delivery and dine-out.
- Swiggy gains **loyal young users** and stronger partner relationships.



Unit Economics of this Business Model IIT (ISM) Dhanbad

Assumptions and Data for Unit Economics

- Total Machines Installed: **20 (in 13 different Hostels)**
- Average Orders per Machine: **50 orders/day**
- Average Order Value (AOV): **₹80 per order**
- Profit Margin (after cost): **35%**
- Machine Cost: **₹5,00,000**
- Institute Deposit: **₹10,000 per machine (one-time)**
- Monthly Operating Cost: **₹50,000 (electricity, refill labor, maintenance)**
- Operational Days per Month: **30**
- Total Monthly Orders = $50 \times 20 \times 30 = 30,000$ orders per month

1. Monthly Revenue (Before Expenses):

Revenue = AOV × Total Monthly Orders = ₹80 × 30,000 = **₹24,00,000**

2. Monthly Gross Profit (Before Fixed Costs):

Gross Profit = 35% of Monthly Revenue = $0.35 \times ₹24,00,000 = ₹8,40,000$

3. Total Monthly Operating Cost:

Electricity + Maintenance + Refilling Labor = **₹50,000/month**

4. Net Monthly Profit (After Costs):

Net Profit = ₹8,40,000 – ₹50,000 = **₹7,90,000**

Expense Category	Cost (₹)
Machine Setup (20 units × ₹5L)	₹1,00,00,000
Institute Deposit (20 units × ₹10K)	₹2,00,000
Installation & Branding	₹1,00,000
IoT Setup & Testing	₹50,000
Total Estimated Investment	₹1,03,50,000

Payback Period (Break-Even):

$$\text{Payback Period} = \frac{\text{Total Investment}}{\text{Net Monthly Profit}}$$
$$\text{Payback Period} = \frac{₹1,03,50,000}{₹7,90,000} = 13.1 \text{ months (~1.1 years)}$$

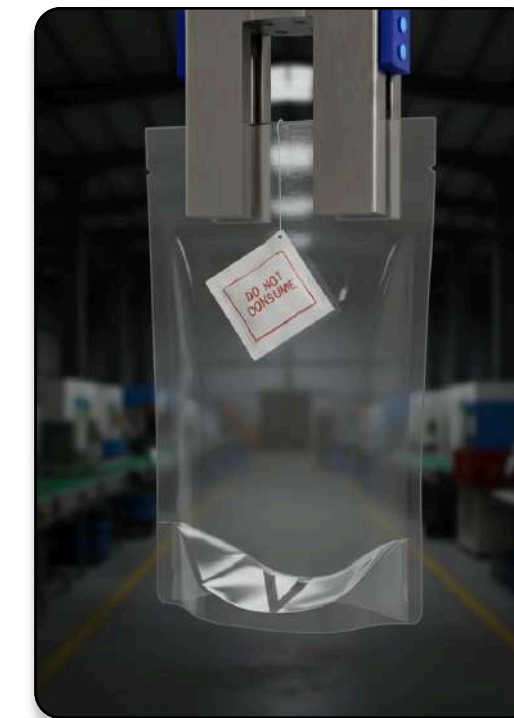
- HydroVending reaches break-even in just over 1 year, even with modest order volume.
- After breakeven, each IIT-scale campus can generate ~₹95 Lakh annual profit.

Possible Pitfalls & Mitigations

1 Desiccant Pouch Mixing with Food During Dispensing

Mitigation :

- Attach the desiccant pouch to the inner wall of the food pack using a heat-resistant string, ensuring it stays outside the food flow when the packet is cut from the bottom.
- Add a colored tag or label on the pouch for easy detection by machine sensors and visual checks during packaging.



2 Power Failure Stopping Vending Operations



Mitigation :

- Equip each HydroVending machine with a UPS backup (15–20 mins capacity) to safely complete any active meal process and store ongoing transaction data.
- Implement an auto-restart recovery system that resumes pending orders once power is restored and sends an instant refund or credit alert to the affected user through the Swiggy app.

3 High Coin Redemption Hurting Restaurant Profitability

Mitigation :

- Set a tier-based redemption limit – for example, coins can cover only up to 30–35% of the bill value, depending on the restaurant type or total bill amount.
- Build a smart redemption system in the app that automatically adjusts limits based on order value.





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➡ Goal / Vision

- To build Swiggy's **first physical touchpoint** inside campuses through HydroVending – delivering convenience, consistency and affordability.
- Over time, these **micro-meal** moments evolve into **macro celebrations** via the Swiggy Coins dine-out ecosystem.

➡ Target Audience

➔ Primary

- Students (18–24 yrs) in hostels looking for **affordable, quick** and **social** food options.

➔ Secondary

- Swiggy partner restaurants seeking **higher visibility** & **consistent student footfall**.

➡ Pilot Region

➔ **IIT (ISM) Dhanbad** : 6,500 students, ideal for controlled launch

➡ Key KPIs (Pilot Targets)

Avg. orders/machine/day

100 +

Student coin redemption rate

>40%

Partner restaurant dine-out growth

+25%

Avg. dine-out savings per group

₹300–500

Swiggy student retention (3 months)

>60%

Avg. Order Value (AOV)

₹80–₹90

➡ Core GTM Strategy : Two-Pronged Approach

1 HydroVending (On-Campus Adoption)

- Install **20–25 HydroVending machines** inside hostels, libraries and canteens.
- Offer **10% Swiggy Coins** on every HydroVending order.
- **App integration**: order tracking, coin balance and quick reorders.
- **Campus ambassadors** to promote usage and run demo events.
- Pricing between **₹60–₹100** to make it affordable for students.

2 Dine-Out (Off-Campus Growth)

- Partner with **10 popular nearby restaurants** for student dine-outs.
- Students can **combine their Swiggy Coins** to get discounts on group bills.
- Restaurants ranked monthly based on **total coins redeemed**.
- **Top 3 restaurants** featured on Swiggy search results (e.g., Pizza Junction – Dhanbad).
- Leaderboard visible inside the app to **motivate** restaurants and students.

➡ Launch Phases

Phase – 1

📍 IIT (ISM) Dhanbad

- Install **20 HydroVending machines** across hostels and canteens.
- Run campus launch with **free first-order** and **ambassador drive**.
- Track daily orders, repeat rate and student feedback for **refinement**.

Phase – 2

📍 Top IITs & Pune

- Expand to **5–6 major campuses** with high hostel density.
- Onboard **15–20 local restaurants** for dine-out integration.
- Launch **Swiggy Coins leaderboard** to boost restaurant participation.

Phase – 3

📍 Tier 1–2 Colleges

- Deploy HydroVending machines across **50+ campuses** nationwide.
- Partner with Swiggy for full in-app “**Campus Zone**” integration.
- Build repeat habit loop between **HydroVending orders** and **dine-outs**.

➡ Activation Channels

1 On-Campus Launch Events

- Demo HydroVending machines in hostels with **free first meal** offers and student ambassador campaigns.

2 Digital Promotions

- In-app Swiggy banners, coin notifications and social media reels showing “**Hot Meals, Instantly.**”

3 Restaurant Co-Branding

- Leaderboard posters, QR coupons, and dine-out discounts at **top Swiggy partner** cafés near campus.